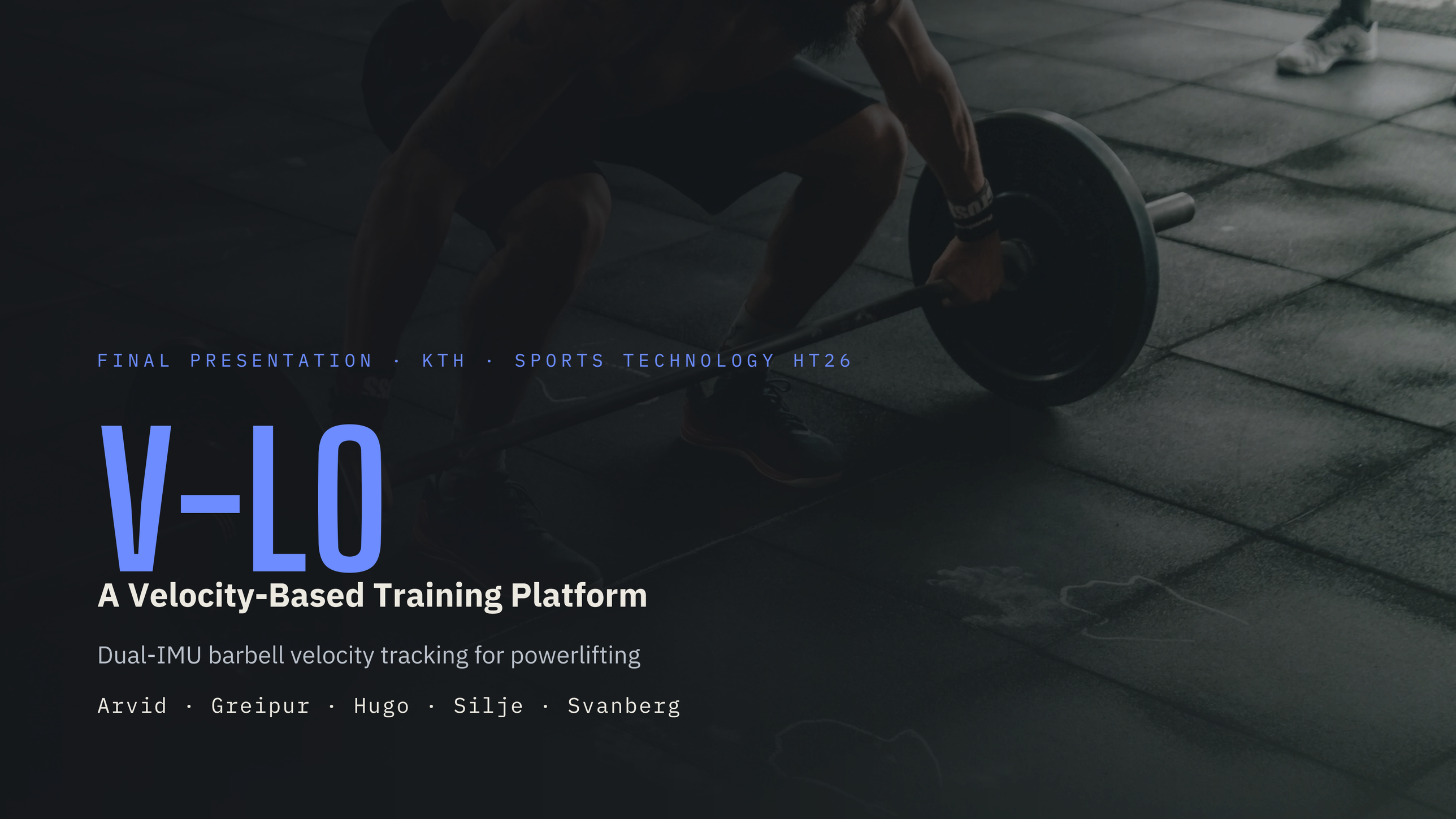


FINAL PRESENTATION · KTH · SPORTS TECHNOLOGY HT26

V-LO

A Velocity-Based Training Platform

Dual-IMU barbell velocity tracking for powerlifting

Arvid · Greipur · Hugo · Silje · Svanberg

01 · PROJECT INTRODUCTION

EVERY REP, MEASURED

V-LO measures how fast the barbell moves on every rep, using two sensors on the bar ends, and turns it into feedback for the next set.

Velocity-based training

Load and effort are guided by bar speed instead of fixed percentages of 1RM.

Intended users

Powerlifters and strength athletes, and the coaches who structure and follow their training. Squat, bench press and deadlift.



01 · PROJECT INTRODUCTION

TRAINING IN VELOCITY ZONES

Absolute strength	Accelerative strength	Strength-speed	Speed-strength	Starting strength
< 0.50	0.50–0.75	0.75–1.00	1.00–1.30	> 1.30
Maximal strength, heaviest loads	Strength with intent to accelerate	Strength moved with speed	Power, moderate loads	Explosiveness, lightest loads

Heavier, slower — Lighter, faster · m/s

The coach picks a zone; V-LO suggests the load that lands the athlete in it, and adjusts it set by set.

Mean velocity of the set, in the same colours as the app. The validation grouped them into three zones by peak velocity.



FROM REP TO RESULT

Demo video · 1:19



V-1

V-10

HOW IT WORKS

THREE STEPS, ZERO LAB COAT.

01

Clip it to the bar

The Movesense Flash snaps into a low-profile sleeve mount. Under 1.0 grams, it
once over Bluetooth.

02

Lift like

02 · PROBLEM STATEMENT

VBT WORKS, BUT IT IS HARD TO ADOPT

Strength gains at least as large as percentage-based loading, at lower loads [1]. Yet coaches report three barriers [3]:

Cost
The most accurate devices are expensive.

Ease of use
Cables and setup get in the way of the lift.

Accuracy
Cheaper wireless sensors drift and suffer from bar rotation.

Existing solution	Strength	Weakness
Linear position transducer (e.g. GymAware)	Most accurate and sensitive [4]	Expensive; the cable can interfere with the lift
Single-IMU devices	Cheaper and wireless	Less accurate [4]: drift and rotation

RESEARCH QUESTION

Can two end-mounted IMUs estimate the mean and peak velocity of each rep with an error below 0.10 m/s?

Small enough that a rep is not placed in the wrong training zone.

03 · TECHNICAL APPROACH

FROM REP TO RESULT

1
Sensors
2× Movesense, accelerometer + gyroscope, 104 Hz

→

2
Web app
Bluetooth in the browser, records the set

→

3
Supabase
PostgreSQL: raw JSON, set details, results

→

4
Azure Function App
Serverless Python, started by a webhook; computes velocity.

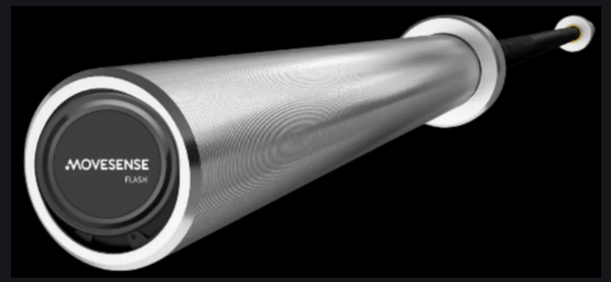
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5
Feedback
Velocity per rep, zone check, next load

Results are saved back to Supabase; the app reads them and shows them a few seconds after the set.

Why Movesense, not a custom board?

A custom IMU would mean our own firmware, Bluetooth stack and enclosure: time taken from the data analysis and the user experience, our core contributions.



03 · TECHNICAL APPROACH

FROM ACCELERATION TO VELOCITY

1 **Orientation**

A Madgwick filter fuses accelerometer and gyroscope, turning the acceleration into a fixed earth frame despite bar tilt [6].

2 **Baseline**

The first 0.5 s, with the bar held still, sets the zero level.

3 **Integration and drift**

Acceleration is integrated to velocity; the drift is modelled as a straight line over the whole recording and removed.

4 **Two sensors, best one**

Both sides are processed; the sensor whose drift fits the line best is used.

5 **Repetitions**

Peaks in the corrected velocity, for the entered number of reps; peak and mean velocity per rep.

04 · USER INTERACTION

ONE PLATFORM, TWO ROLES

Coach plans

Assigns workouts where each lift gets a target velocity zone instead of a fixed weight, and follows every athlete’s results.

Athlete records

Sees the day’s workout and a suggested starting load from their own history, then records the set from the browser.

Feedback in seconds

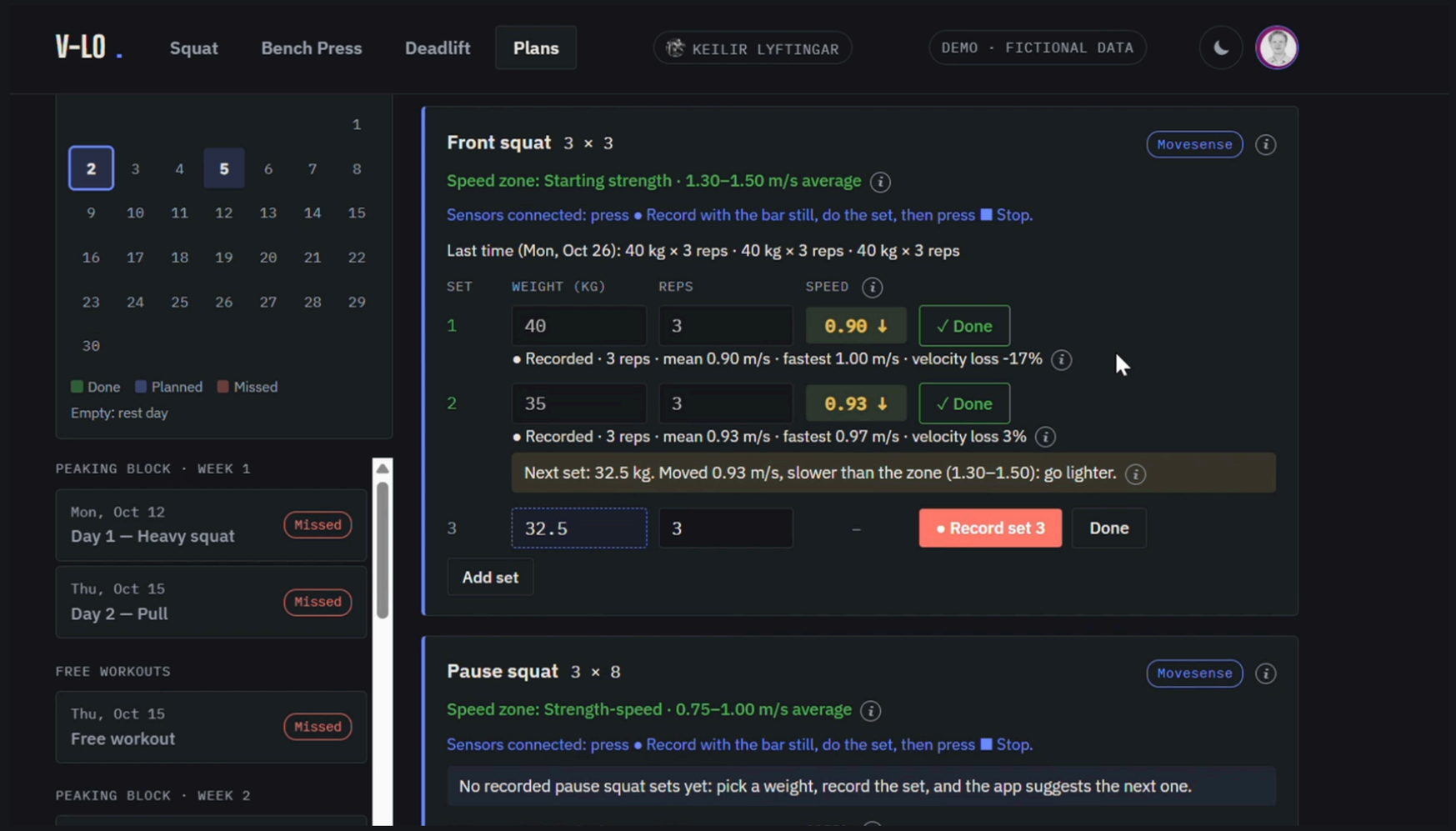
Mean and peak velocity of every rep, whether the set hit the zone, and a suggested load for the next set.

Team space

Teams of coaches and athletes, with a leaderboard of the best performing sets in each velocity zone.

Recording a set: `Sensors on, connect` → `● Record, bar still for a moment` → `Perform Set` → `■ Stop` → `Results and feedback`

04 · USER INTERACTION



05 · RESULTS

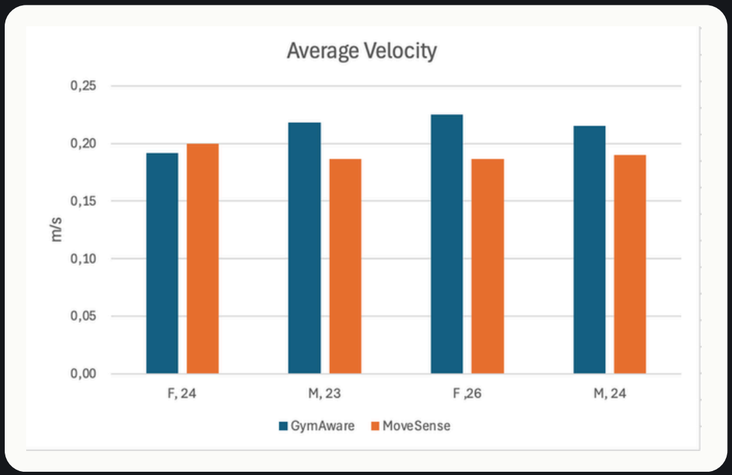
ACCURATE WHERE POWERLIFTERS TRAIN

4 participants, 6 squats in 3 velocity zones, recorded at the same time with V-LO and a GymAware RS. Mean absolute error ± SD, in m/s:

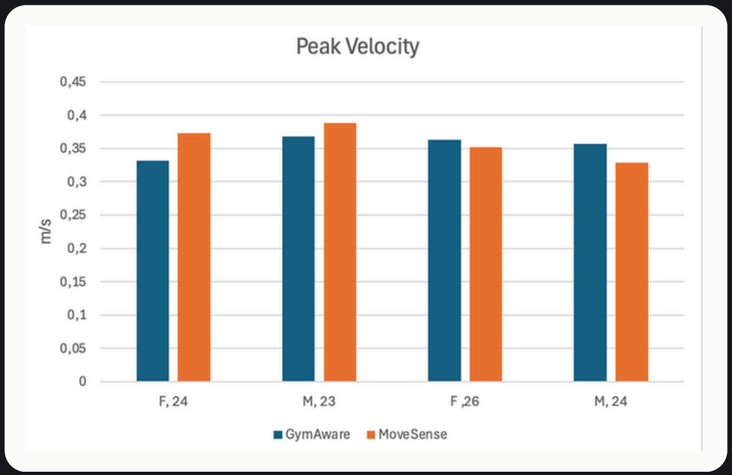
Zone (peak velocity)	Mean velocity	Peak velocity	Below 0.10 m/s?
Absolute strength (< 0.5)	0.03 ± 0.02	0.04 ± 0.02	Yes
Speed strength (0.5–1.3)	0.05 ± 0.16	0.06 ± 0.03	Yes
Starting strength (> 1.3)	0.33 ± 0.17	0.21 ± 0.13	No

Peak velocities above 1.3 m/s only occur with very light loads, rarely used in powerlifting.

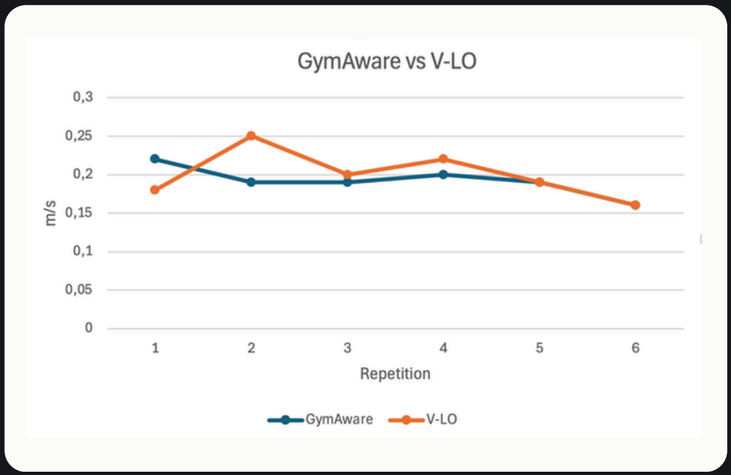
ABSOLUTE STRENGTH, PARTICIPANT BY PARTICIPANT



Mean velocity per participant, absolute strength



Peak velocity per participan, absolute strengtht



Rep by rep, one set for one participant

V-LO follows the GymAware within a few hundredths of a metre per second.

05 · RESULTS

FEEDBACK BEFORE THE NEXT SET

5.8 s
Median processing time

0.9–7.5 s
Range over 7 recorded sets

Measured from the recording being stored until its results were available; A preliminary test.

CONCLUSION

ACCURATE ENOUGH FOR POWERLIFTING LOADS

Two end-mounted Movesense IMUs stayed below 0.10 m/s for low and moderate velocities, with feedback in seconds. Fast, light lifts need better drift handling.

Limitations

- Four participants, squats only
- Sensitive to integration drift
- The rep count is entered before each set

Future work

- Better drift handling for fast lifts
- Bench press and deadlift
- A larger study with powerlifters
- Combining both sensors, e.g. left–right balance
- Automatic rep detection, no count to enter
- Estimating 1RM from the load–velocity profile
- Making the website choose the weight for each set intelligently

06 · PROJECT METHODOLOGY

TWO TEAMS, ONE PIPELINE

A data team built the velocity algorithm and the validation; a product team built the platform, database and cloud processing. Work was tracked as GitHub issues and sub-issues.

What went well

- Two teams working in sprints: algorithm and platform in parallel
- Off-the-shelf sensors: no firmware or hardware work
- Issues and sub-issues kept everyone’s tasks visible
- An early end-to-end pipeline, from recording to feedback

What we would do differently

- Test with a real barbell and both sensors earlier
- Agree on the recording protocol and data format between the teams from the start
- Integrate the algorithm into the platform continuously
- Write down the MVP scope and priorities in week one

V-LO

Thank you. Questions?

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[6] Madgwick, Harrison & Vaidyanathan, “Estimation of IMU and MARG orientation using a gradient descent algorithm,” IEEE ICORR, Zurich, 2011, pp. 1–7.

[7] GymAware, “GymAware | Gold Standard Velocity Based Training Equipment,” gymaware.com (retrieved 19-10-2026).